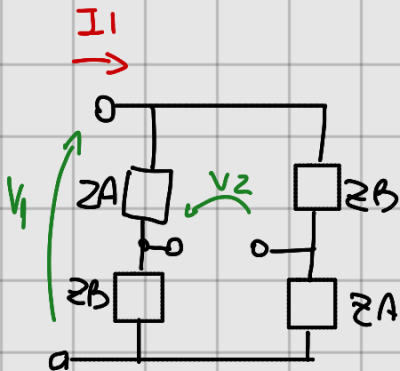


OBTENGO LATTICE T

$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$$

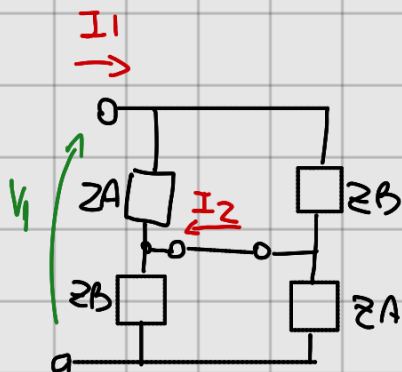


$$V_2 = V_1 \cdot \frac{Z_B}{Z_A + Z_B} - V_1 \cdot \frac{Z_A}{Z_A + Z_B}$$

$$V_2 = V_1 \cdot \frac{Z_B - Z_A}{Z_A + Z_B}$$

$$A = \frac{Z_A + Z_B}{Z_B - Z_A}$$

$$B = \left. \frac{V_1}{-I_2} \right|_{V_1=0}$$



$$I_2 = I_{Z_B} - I_{Z_A}$$

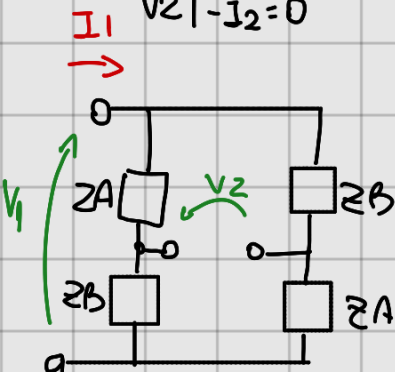
$$I_2 = \frac{V_1}{2} \cdot \frac{1}{Z_B} - \frac{V_1}{2} \cdot \frac{1}{Z_A}$$

$$I_2 = V_1 \cdot \left(\frac{1}{2Z_B} - \frac{1}{2Z_A} \right)$$

$$-I_2 = V_1 \cdot \left(\frac{Z_B - Z_A}{2Z_A Z_B} \right)$$

$$B = \frac{2 \cdot Z_A Z_B}{Z_B - Z_A}$$

$$C = \left. \frac{I_1}{V_2} \right|_{-I_2=0}$$

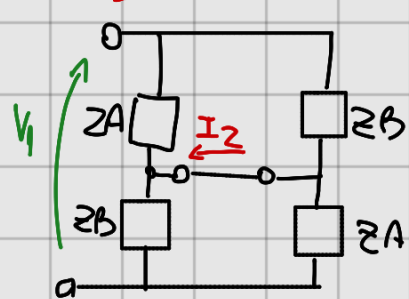


$$V_2 = \frac{I_1}{2} \cdot Z_B - \frac{I_1}{2} \cdot Z_A$$

$$V_2 = I_1 \cdot \frac{Z_B - Z_A}{2}$$

$$C = \frac{2}{Z_B - Z_A}$$

$$D = \frac{I_1}{-I_2} \bigg|_{V_2=0}$$



$$I_2 = I_{ZB} - I_{ZA}$$

$$I_2 = V_1 \cdot \left(\frac{1}{2ZB} - \frac{1}{2ZA} \right)$$

$$I_1 = \frac{V_1}{2 \cdot (ZA/ZB)} = \frac{V_1}{2} \cdot \frac{ZA+ZB}{ZA \cdot ZB}$$

$$V_1 = I_1 \cdot 2 \cdot \frac{ZA \cdot ZB}{ZA+ZB}$$

$$I_2 = I_1 \cdot \frac{\cancel{ZA \cdot ZB}}{ZA+ZB} \cdot \cancel{2} \cdot \left(\frac{\cancel{2} \cancel{ZA} - \cancel{2} \cancel{ZB}}{\cancel{4} \cancel{ZA \cdot ZB}} \right) = I_1 \cdot \frac{ZA - ZB}{ZA+ZB}$$

$$D = \frac{ZA+ZB}{ZB-ZA}$$

COMPARE AMBAS MATRICES

$$T_{LAT} = \begin{pmatrix} \frac{ZA+ZB}{ZB-ZA} & \frac{2 \cdot ZA \cdot ZB}{ZB-ZA} \\ \frac{2}{ZB-ZA} & \frac{ZA+ZB}{ZB-ZA} \end{pmatrix}$$

$$\textcircled{I} \quad \frac{ZA+ZB}{ZB-ZA} = \frac{S^2+3}{S^2+1}$$

$$\textcircled{II} \quad \frac{2 \cdot ZA \cdot ZB}{ZB-ZA} = 2 \cdot \frac{2S^2+4}{S^3+S}$$

$$T_T = \begin{pmatrix} \frac{S^2+3}{S^2+1} & \frac{4S^2+8}{S^3+S} \\ \frac{S}{S^2+1} & \frac{S^2+3}{S^2+1} \end{pmatrix}$$

$$\textcircled{III} \quad \frac{2}{ZB-ZA} = \frac{S}{S^2+1}$$

II/III

$$Z_A \cdot Z_B = 2 \cdot \frac{2s^2+4}{s^2+8} \cdot \frac{s^2+1}{s}$$

$$Z_A \cdot Z_B = \frac{4s^2+8}{s^2} \quad \text{IV}$$

I/III

$$\frac{Z_A+Z_B}{2} = \frac{s^2+3}{s^2+1} \cdot \frac{s^2+1}{s}$$

$$Z_A+Z_B = 2 \cdot \frac{s^2+3}{s} \quad \text{V}$$

Opciones...

$$L/C: \frac{s \cdot \frac{1}{C_1}}{s^2 + 1/C_1}$$

$$R/C: \frac{1/C}{s + 1/R}$$

$$L+C: \frac{s^2 \cdot L + 1/C}{s} \quad R+C: \frac{\frac{1}{C} + R}{s}$$

Determino...

$$Z_A: \text{---} \frac{L}{C_1} \text{---} : \frac{s^2 L + 1/C_1}{s}$$

$$Z_B: \text{---} \frac{C_2}{C_2} \text{---} : \frac{1/C_2}{s}$$

Retomando ecuaciones.

$$\text{IV} \quad Z_A \cdot Z_B = \frac{s^2 \cdot L + 1/C_1}{s} \cdot \frac{1/C_2}{s} = \frac{s^2 \cdot L \cdot (\frac{1}{C_2} + R) + \frac{1}{C_1} \cdot \frac{1}{C_2}}{s^2}$$

$$\frac{s^2 \cdot L \cdot \frac{1}{C_2} + \frac{1}{C_1} \cdot \frac{1}{C_2}}{s^2} = \frac{4s^2+8}{s^2}$$

$$L \cdot \frac{1}{C_2} = 4 \text{ VI}$$

$$\frac{1}{C_1} \cdot \frac{1}{C_2} = 8 \text{ VII}$$

$$(*) \quad Z_A + Z_B = \frac{S^2 L + 1/C_1}{S} + \frac{1/C_2}{S} = \frac{S^2 L + \frac{1}{C_1} + \frac{1}{C_2}}{S}$$

$$\frac{S^2 L + \frac{1}{C_1} + \frac{1}{C_2}}{S} = \frac{2S^2 + 6}{S}$$

$$\bullet L = 2 \text{ VIII}$$

$$\bullet \frac{1}{C_1} + \frac{1}{C_2} = 6 \text{ IX}$$

Operando...

$$\text{VIII} \quad L = 2 \quad \frac{\text{VII}}{\text{VI}} \quad \frac{1}{C_1} \cdot \frac{1}{L} = 2 \quad C_1 = \frac{1}{4}$$

$$\text{IX} \quad \frac{1}{C_2} + \frac{1}{C_1} = 6 \quad C_2 = 1/2$$

$$\text{VII, VI} \quad L \cdot \left(\frac{1}{C_2}\right) = 4 \checkmark$$

$$\frac{1}{C_1} \cdot \frac{1}{C_2} = 8 \checkmark$$